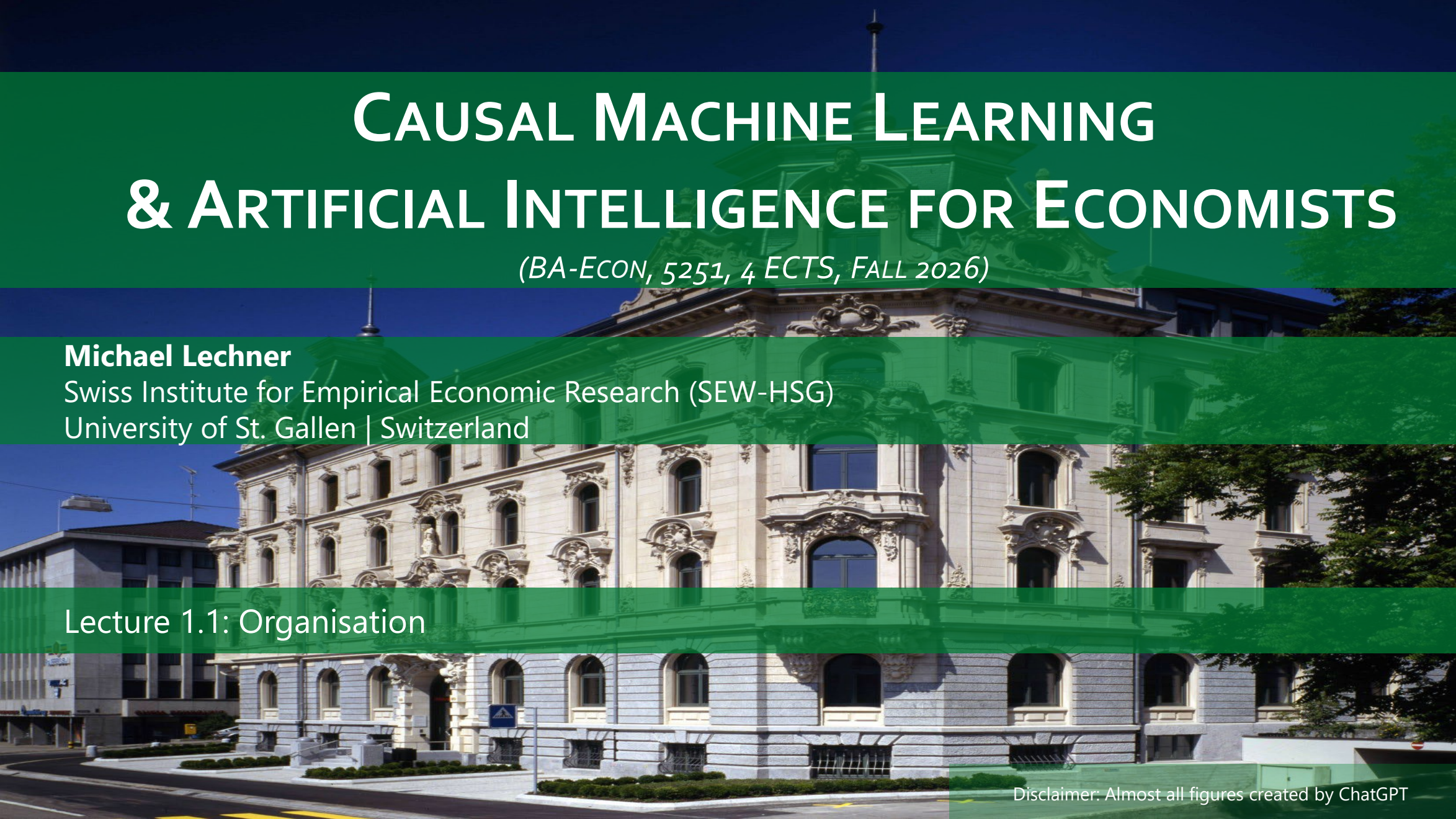
Causal Machine Learning & Artificial Intelligence for Economists

(BA-Econ, 5251, 4 ECTS, Fall 2026)

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University of St. Gallen | Switzerland

Lecture 1.1: Organisation



Disclaimer: Almost all figures created by ChatGPT

Double Machine Learning and **Causal Forests**;



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Content

Introduction to intersection of **Causal Inference**, **Machine Learning**, & **Artificial Intelligence** (data-driven decision making)

Topics of the lectures

- Prediction & Causation (week 1)
- Machine learning for prediction (**ML,** week 2)
- Machine learning for the estimation of causal effects (**C**ausal**ML,** week 3 & 4)
- Using causal heterogeneity obtained from **CML** to improve decision making (Optimal Policy, **AI,** week 5)
- Own empirical (group) projects using all these tools (week 6)



Goals

After successfully completing the course, students should be able to:

- explain the difference between **prediction** & **causal** questions
- assess simple **empirical research designs** & understand their role for **causal interpretations of** empirical estimates
- apply basic methods of **machine learning for prediction** & discuss their advantages & limitations relative to classical econometric approaches
- understand the central concepts of **causal machine learning**, especially for estimating average & heterogeneous effects
- conceptually understand & competently use methods such as **Double Machine Learning** & **Causal Forests**
- understand data-driven approaches to **decision making** & **optimal policy assignment**, in particular policy trees
- conduct an empirical project using appropriate data, methods, interpretations, & present the results in a scientifically acceptable way



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Why Python?

Popular in industry & scientific community

- Standard in the industry using machine learning & causal machine learning
  - Course & CML in general uses many components/packages from the Python eco-system
- Rapidly increased popularity in the research community

Huge community leads to higher quality of LLM tools to assist coding

Python skills (general programming language) are also very useful outside data analysis

Python skills in quantitative methods increases employability of students

The TIOBE Programming Community index is an indicator of the popularity of programming languages. The index is updated once a month. The ratings are based on the number of skilled engineers world-wide, courses and third party vendors. Popular web sites Google, Amazon, Wikipedia, Bing and more than 20 others are used to calculate the ratings. It is important to note that the TIOBE index is not about the *best* programming language or the language in which *most lines of code* have been written.

The index can be used to check whether your programming skills are still up to date or to make a strategic decision about what programming language should be adopted when starting to build a new software system. The definition of the TIOBE index can be found here.

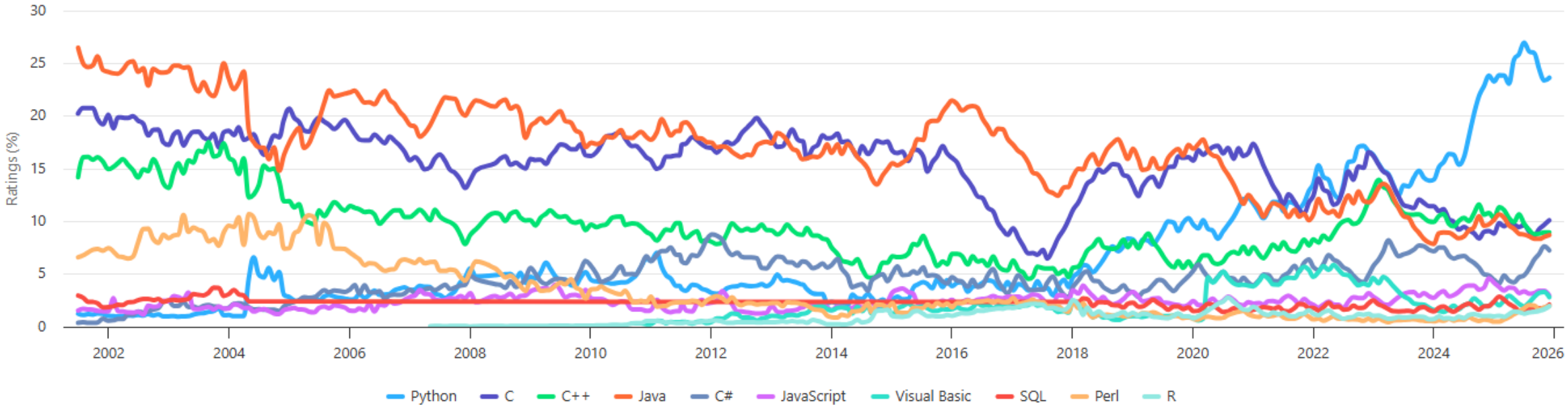
Dec 2025	Dec 2024	Change	Programming Language	Ratings	Change
1	1		Python	23.64%	-0.21%
2	4	↑	C	10.11%	+1.01%
3	2	↓	C++	8.95%	-1.87%
4	3	↓	Java	8.70%	-1.02%
5	5		C#	7.26%	+2.39%
6	6		JavaScript	2.96%	-1.66%
10	16	⇈	R	1.96%	+0.91%
11	11		Delphi/Object Pascal	1.91%	+0.48%
12	10	↓	Fortran	1.60%	-0.18%
13	15	↑	MATLAB	1.52%	+0.43%

TIOBE Index – TIOBE (www.tiobe.com/tiobe-index/)



TIOBE Programming Community Index

Source: www.tiobe.com



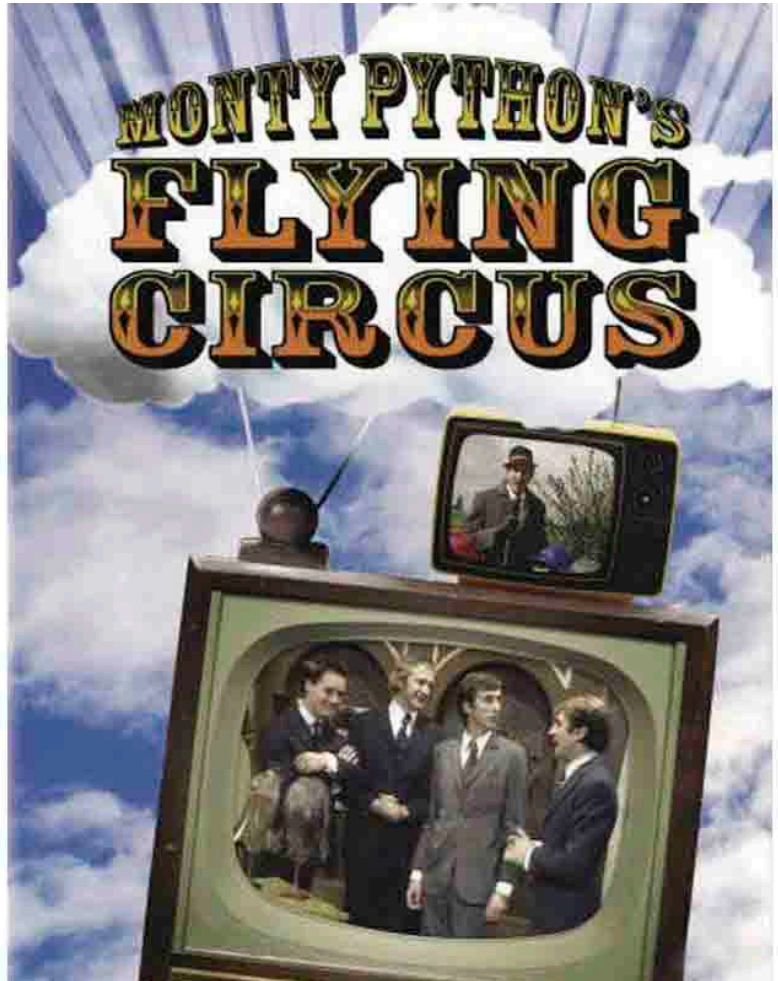
University of St.Gallen
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Empirical Economic Research

python™



Fun fact: Where does the name *Python* come from?









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Sessions | 1

After the break, we meet every Monday, 8:15-12:00, C 83-1235

Each session has 2 parts

- Lecture: Theoretical concepts, methodology, etc.
- Applied work: Students work (more or less) in groups on empirical problem sets (guided by lecturer)

Slides & other material are available on Canvas



Organizational issues | 2

The default IDE is Spyder (*IDE: Integrated Development Environment*)

- It must be installed **on your own computer** & running **before the 1st session**
  - Installing it using the *Anaconda Distribution* is recommended (https://www.anaconda.com/products/distribution)
    - Anaconda is available for all major operating systems
  - After installing Anaconda, start *Anaconda Navigator*
    → now Spyder can be launched
  - **OR:** *conda install spyder* (in Anaconda terminal of environment)

Other GUI may be used (not supported by us)

- Avoid notebooks

Installation

1. Download the Anaconda installer.
2. Go to your Downloads folder and double-click the installer to launch. To prevent permission errors, do not launch the installer from the Favorites folder.

> **Note**
> If you encounter issues during installation, temporarily disable your anti-virus software during install, then re-enable it after the installation concludes. If you installed for all users, uninstall Anaconda and re-install it for your user only.

3. Click **Next**.
4. Read the licensing terms and click **I Agree**.
5. It is recommended that you install for **Just Me**, which will install Anaconda Distribution to just the current user account. Only select an install for **All Users** if you need to install for all users' accounts on the computer (which requires Windows Administrator privileges).
6. Click **Next**.
7. Select a destination folder to install Anaconda and click **Next**. Install Anaconda to a directory path that does not contain spaces or unicode characters. For more information on destination folders, see the FAQ.

> **Caution**
> Do not install as Administrator unless admin privileges are required.



Grades

Group presentation of empirical projects (50% of final grade) in last lecture

- Tasks will be distributed in the beginning of the last lecture, **Dec., 14**
- Each group solves problems allocated to them & prepares a presentation
- Presentations will be held in this lecture as well (graded)
- Group members are to be decided not later than after 3rd lecture (***Nov., 23, 15:00***)
  - If you do not belong to a group, you are a one-person group (designated group-leader should send an email to Federica.Mascolo@unisg.ch & me with names of all group members, & all group members in cc)
- This group exam will be under immense time pressure ➔ it is important that groups use the previous 5 group work sessions to learn how to efficiently collaborate

Oral exam on Thursday, **Dec., 17** (50% of final grade)



Rules for group work (including graded group exercise)

Groups may have up to 4 members

- *All group members must agree to all group members*

Group memberships may change until lecture 3

You are free to use any AI you like

- Graded exam: Contact with humans is prohibited & will be monitored (& will be useless)



Lecturer is available for discussions & help during group work

- limited in graded group work

You should always ***install & test*** the required packages ***before the lecture***

- ***Required (additional) packages are always shown on 1st slide of each lecture***



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You

Which HSG programme?

Knowledge about empirical methods?

Experience with coding?

- Python
- R
- MATLAB
- Stata
- C++

Expectations of this course?

Please briefly introduce yourself



Our team



Michael Lechner

- Professor of Econometrics & Director of SEW-HSG
- Educated in economics in Heidelberg, LSE, Mannheim (PhD), Harvard (Post doc)
- Research interests in *causal machine learning*, *labour economics*, *sports economics*, *evidence-based policy*, …
- www.michael-lechner.eu, www.researchgate.net/profile/Michael_Lechner, www.sew.unisg.ch/en/lehrstuehle/empirische-wirtschaftsforschung



Federica Mascolo (organizational matters)

- 4th year PhD students (GPEF)
- Thesis on topics in causal machine learning



Data & method consulting at HSG

4 consultants support students in their empirical studies and thesis projects



Johanna Kutz
Quantitative methods
(Econ focus) & ML



Emanuel de Bellis
Quantitative methods
(Business focus)



Hugo Bodory
Programming & IT,
Quantitative methods



Daniel Dietrich
Qualitative methods

More information about the consultants & a contact form on

http://www.unisg.ch/data-consulting



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Next lecture

Prediction & Causality